

Test Setup and Performance Evaluation of an Infrared Sensor

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Motivation & Context

Satellite-based infrared sensor technology like from the Munich-Based startup OroraTech GmbH enables large-scale, continuous wildfire monitoring [1], but typically requires specialized and expensive space sensors. This is where this student research project comes in: it investigates whether an industrial thermal imaging sensor, the Teledyne FLIR Boson+ bolometer, achieves the necessary performance characteristics for early and reliable forest fire detection in low earth orbit (LEO). This project is a part of the SeeSat e.V. ERWIN mission. Using industrial LWIR sensors could reduce mission cost and broaden continuous wildfire monitoring coverage from LEO.

Previous Work

Previous scientific Work already determined mission parameters and undertook necessary atmospheric simulations using MODTRAN [2], identifying the atmospheric transmission coefficients for the situation.

Imager Specs

Param	Value	Description
Wavelength-Range	8 – 14 μm	LWIR-Spectral-Range
NETD	40 mK, 50 mK, 60 mK	Detector-Noise
Pixel Pitch	12 μm	Pixel Pitch of the Detector

Mission Environment

Param	Value	Description
GSD	197 m	Ground Sampling Distance for a single Pixel
Distance	600 km	Common Orbit for CubeSats
Optical Transmission	0.5	Estimation for a 5 Lens Design
Aperture Diameter	0.06 m	Estimation
Detector Efficiency	0.8	Estimation

MODTRAN Data

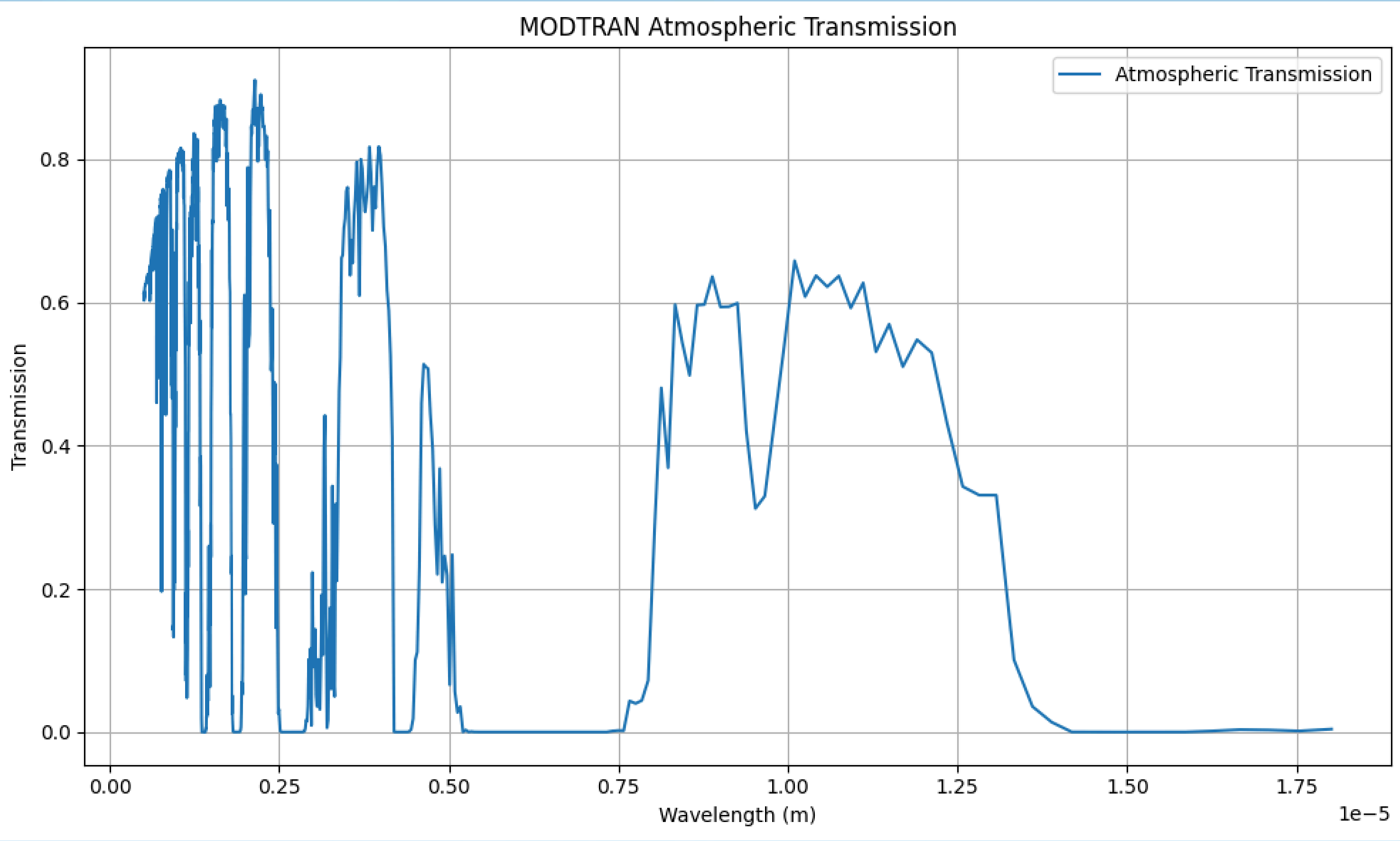


Figure 1: Atmospheric Transmission

MODTRAN runs identified the relevant 8–14 μm atmospheric transmission window.

Simulation

Using the mission parameters, MODTRAN coefficients and Planck’s law of radiation [3], the total Energy reaching a single pixel of the imager can be calculated.

$$P_{ow} = A_{surface} \cdot \left[\sum_{n=1}^N \tau_n \cdot \int_{\lambda_n}^{\lambda_{n+1}} S_{\lambda}(\lambda, T) d\lambda \right] \cdot \pi \sin^2(\theta_{ap}) \cdot a$$

Figure 2: Universal Pixel Power Equation

where $S_{\lambda}(\lambda, T)$ is Planck’s law of radiation, τ_n is the atmospheric transmission for a wavelength band, $A_{surface}$ is the area on the surface seen by a single pixel, λ_n are the band edges, θ_{ap} is the aperture half-angle and a are further factors depending on the type of radiation examined:

Surface: Radiation stemming directly the ground.
Albedo:Diffractional sun reflections from earth’s surface.
Glint: Strong specular sun reflections.

This equation assumes Lambertian emission [4] (surface radiates equally in all directions) and the imager looking directly at the ground.

Smallest Detectable Fire

Determining the smallest detectable fire requires:

- The power difference of the burning ground and normal ground being higher than the noise equivalent power (NEP) of the detector.
- The power being larger than the total power of the ground in addition to reflective noise.

Power Difference

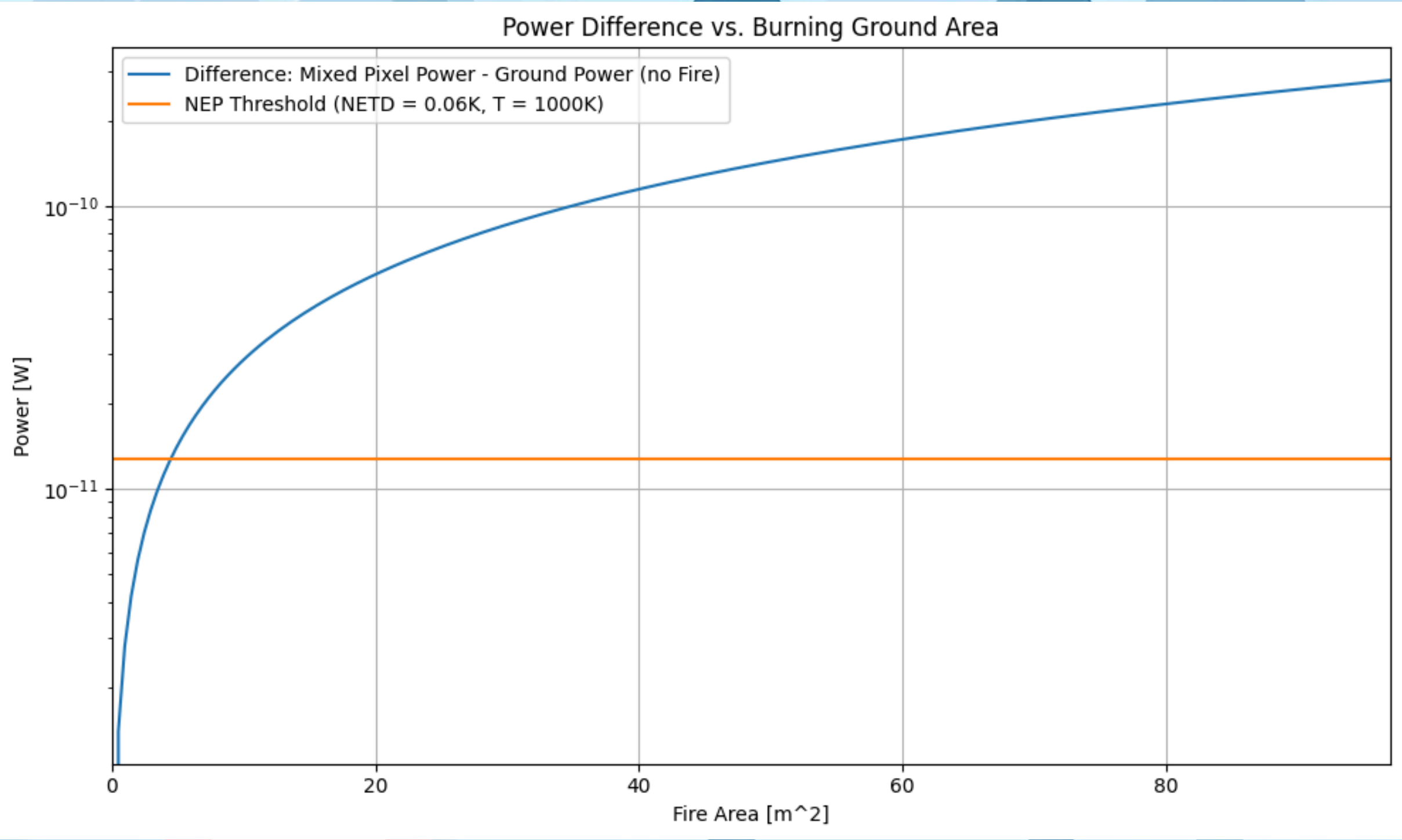


Figure 3: Fire Power Difference Simulation

Reflective Noise

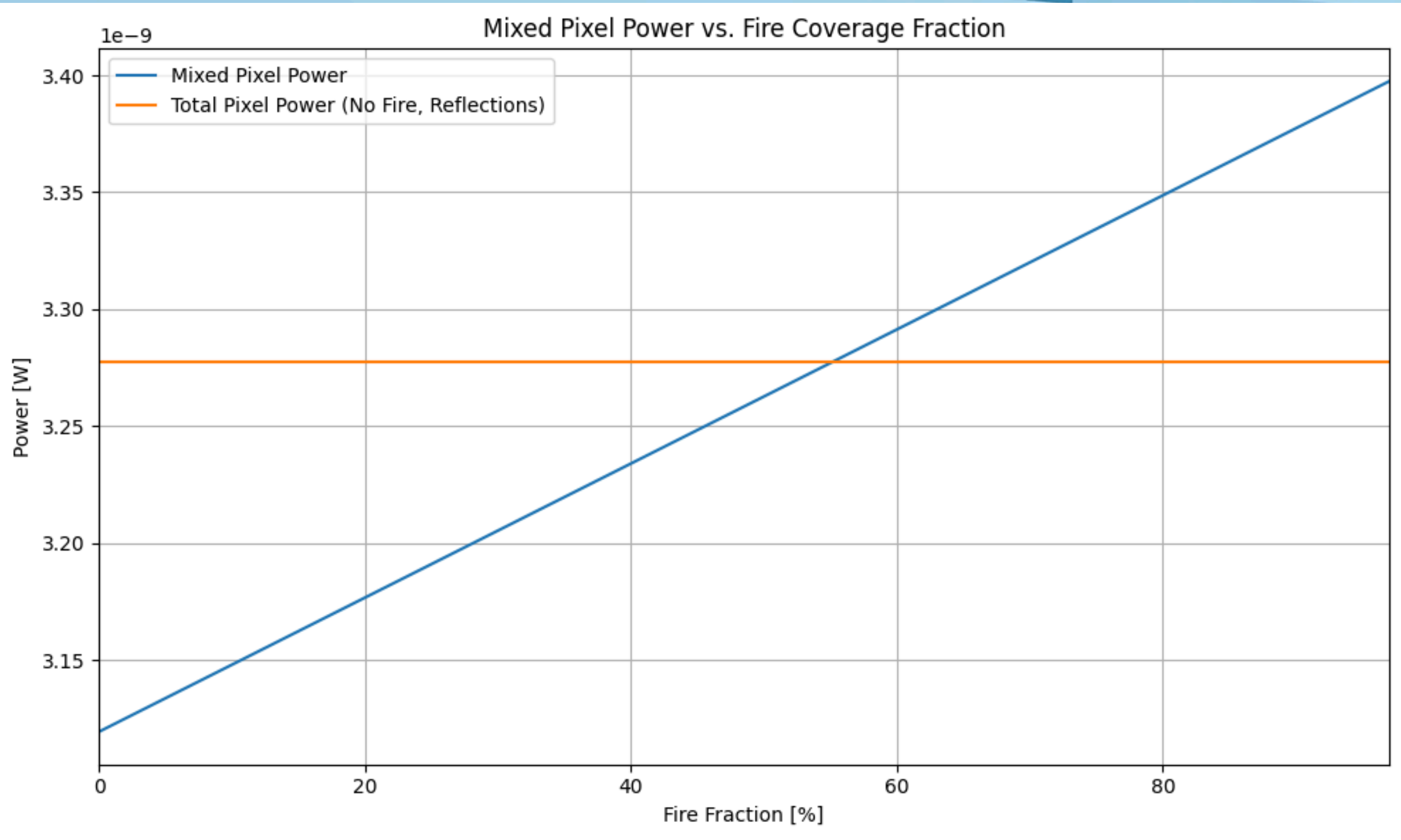


Figure 4: Total Pixel Power Simulation

The Plots show that the criteria is met, for fires of sizes $55.6m^2$ and up. This is deemed small enough for early wildfire detection and as such the ERWIN mission.

Outlook & Next Steps

The next steps include verifying the calculations and simulation results with tests. To achieve this, the noise equivalent temperature difference (NETD) of the imager has to be determined and compared with the value from the data sheet, as well as the absolute temperature accuracy and the temperature dependency on the detector performance.

References

[1] Ororatech, “Ororatech Website.” [Online]. Available: <https://www.ororatech.com/>

[2] Spectral Sciences Inc., “MODTRAN® - Radiative Transfer Software.” 2025.

[3] P. A. Tipler and G. Mosca, *Physik für Wissenschaftler und Ingenieure*, 7th ed. Berlin Heidelberg: Springer Spektrum, 2015. doi: 10.1007/978-3-642-54166-7.

[4] M. Bukshtab, *Photometry, Radiometry, and Measurements of Optical Losses*, vol. 209. in Springer Series in Optical Sciences, vol. 209. Singapore: Springer, 2019. doi: 10.1007/978-981-10-7745-6.

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